Algorithm: Dijkstra's Shortest Path Algorithm using Priority Queue

Input:

- Number of locations (vertices).
- Roads connecting locations (edges).
- Distance and travel time for each road.
- Source location.
- Destination location.

Output:

- Fastest available route.
- Total distance.
- Total travel time.

Steps

1. Start.
2. Create a weighted graph using an adjacency list.
3. Represent each location as a vertex and each road as an edge.
4. Store the distance and travel time associated with every road.
5. Initialize the travel time of the source vertex to 0 and all other vertices to infinity.
6. Insert the source vertex into a priority queue.
7. Extract the vertex with the minimum travel time from the priority queue.
8. Visit all its adjacent vertices.
9. Calculate the new travel time for each adjacent vertex.
10. If the new travel time is less than the previously recorded travel time:
    - Update the travel time.
    - Update the total distance.
    - Store the parent vertex for route reconstruction.
    - Insert the updated vertex into the priority queue.
11. Repeat steps 7–10 until the priority queue becomes empty.
12. If the destination is unreachable, display "No available route".
13. Otherwise, reconstruct the fastest route using parent information.
14. Display the route, total distance and travel time.
15. For road closure, delete the corresponding edge from the graph.
16. For road reopening, add the edge back to the graph.
17. Re-run Dijkstra's algorithm to calculate the updated fastest route.
18. Stop.